

First Name:

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Answer

the university of
OKLAHOMA

School of Computer Science
CS 2614: Computer Organization
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Quiz - 6

The following register transfers are to be executed in the system shown.

For each transfer, specify:

- (1) the binary value that must be applied to bus select inputs S_2 , S_1 , and S_0 ;
- (2) the register whose LD control input must be active (if any);
- (3) a memory read or write operation (if needed); and
- (4) the operation in the adder and logic circuit (if any).

(a) $PC \leftarrow DR$

(b) $M[AR] \leftarrow PC$

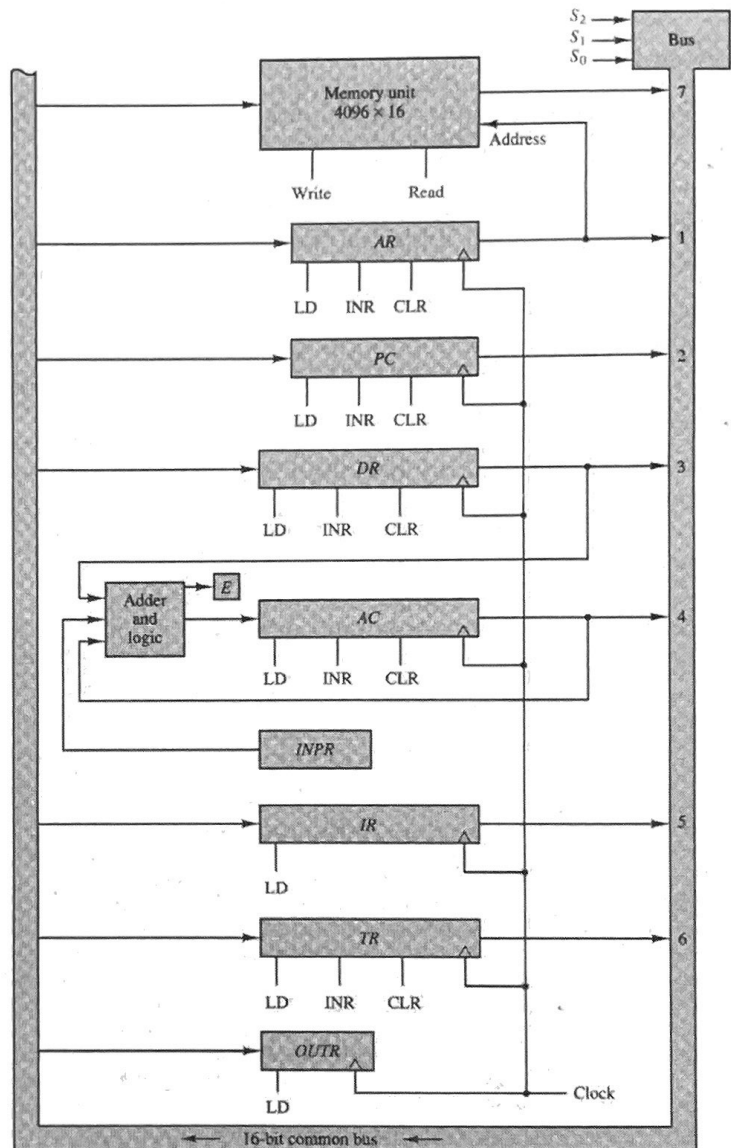


Figure 5-4 Basic computer registers connected to a common bus.

Ans:

	$S_2 S_1 S_0$	Load (LD) of register	Memory operation	Adder and logic operation
$PC \leftarrow DR$	0 1 1	PC		
$M[AR] \leftarrow PC$	0 1 0		Write	